

MusicTongue: A Tongue-Training Exergame Design for the Older Adults

Chia-Hsu Huang
National Taipei University of
Technology, Taipei Tech
mxair95000@gmail.com

Kuei-Ru Chou
Taipei Medical University
kueiru@tmu.edu.tw

Ko-Chiu Wu
National Taipei University of
Technology, Taipei Tech
kochiuwu@mail.ntut.edu.tw

ABSTRACT

Swallowing ability declines with aging, which may cause problems such as choking in eating and difficulty in oral expression, reducing the quality of older adult's life. This study collaborated with medical experts and older adults to develop an oral-training exergame app—MusicTongue. The game design is similar to "Taiko no Tatsujin." Players can follow the music song they love and train the tongue with music beats. The game can recognize the player's movements through the Front-facing camera. After the game is finished, the game will show the training record. In the final iteration, we found that most elders can operate the game smoothly, leading to a greater willingness to perform oral-training exercises.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**; • **User studies**;

KEYWORDS

Exergame, Tongue-Training, Image-Recognition, Human-Centered Design

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1 INTRODUCTION

About 14%~16% of adults over 60 years have swallowing difficulty [1], the percentage is higher in the long-term care facilities, reaching 70% [2]. As a result of natural aging or brain damage caused by accidents may result in dysphagia [3, 4], which may increase the risk of aspiration pneumonia; this affects the life quality of older adults [5]. Oral training can improve the tongue's strength and coordination, maintain the patient's oral function [6]. Existing research on oral training games is more concerned with technical feasibility and efficacy than user experience, and the design process lacks the participation of older people. Therefore, this study involved experts and older adults in the design process, finally creating an

oral-training game—MusicTongue. The training method is based on the "Oral Care Program" [7], which includes a standard series of oral-training exercises that can be performed at home by the older adult; the training will strengthen the muscles related to mouth function [8].

1.1 Related Works

Exergames have been considered an effective solution to promote the motivation of exercise [9]. The existing tongue-training game can be divided into contact and non-contact types (Table 1). The contact type depends on the IOPI trainer [10], textile pressure sensors [11], or intra-oral joystick [12], while the non-contact type uses image recognition to detect mouth movements [13, 14]. Most designs use laptops to operate; it is heavy and not user-friendly. Although IOPI trainers or intra-oral joysticks are accurate, they require expert assistance to run the game, making it difficult for older adults to perform the training themselves. Besides, the game theme also affects the older adults' willingness to use it [15]; existing studies lack research on the user experiences of older adults, so it is not easy to know whether the game is of interest to older adults.

1.2 Limitations

This research is a pilot study; the training protocol used in-game only included tongue exercise. The game does not include the cheek, throat, and palate training. Therefore, the training method in the game can only train oral preparation phases and oral transit phases of the four stages of swallowing (1).

2 RESEARCH PROCESS

First, we interviewed experts to develop training movements for the game, then interviewed older adults to find the potential game themes from participants' leisure interests. After the first version of the prototype design is completed, four iterations begin. The technical prototype is implemented after the second iteration; we collected images of older adults' tongue training postures in the implementation phase to build a recognition dataset for TensorFlow Lite (TF-Lite), then created a technical prototype. Finally, 36 older adults participated in the technical prototype test.

2.1 Study 01: Expert Interview and Training Protocol Design

We interviewed a total of five experts from E1 to E5 (Table 2); the nursing and speech therapy-related experts E1~E3 designed the training protocol used in the game, which was based on the tongue exercise movement in "Oral Care Program" [7]. The protocol included five kinds of training action A1, A2-1, A2-2, A3, and A4 (Table 3); the training period was eight weeks, with at least three

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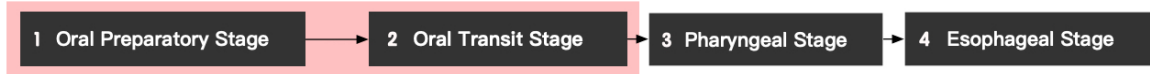
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Table 1: Existing Research in Tongue-training Game

Type	Name & Year	Technical Approach	Platform
Non-Contact	SITA, 2013 [14]	Kinect	PC
Non-Contact	Tongue Drive System, 2014 [16]	Magnetic Sensor	PC
Non-Contact	Squachu, 2018 [13]	Webcam	PC
Contact	Tongue-strengthening Training Tablet Game, 2020 [10]	IOPI	Tablet
Contact	Intra-oral Joystick, 2019 [12]	Intra-oral joystick	PC

**Figure 1: Four Phases of Swallowing****Table 2: Expert List**

Expert Number	Occupation
E1	Nurse
E2	Nursing Expert/Scholar
E3	Speech Therapist/Scholar
E4	Medical Aids Designer/Scholar
E5	Image Recognition Engineer

Table 3: Training Protocol A1 ~ A4

Num	Instruction	Definition to recognition model
A1	"Stick the tongue out of the mouth as far as possible, then retract it, repeat for times."	B1 Spit the tongue down→ B2 Close the mouth.
A2-1	"Extend the tongue to the left; extend the tongue to the right; repeat ten times. "	B3-1 Extend the tongue right→ B2 Close the mouth
A2-2		B3-2 Extend the tongue left→ B2 Close the mouth
A3	"Open your mouth as wide as you can, use the tongue to circle the corners of the mouth, repeat ten times. "	B4 Open the mouth as wide as you can→ B1 Spit the tongue down→ B3-2 Extend the tongue left→ B5 Extend the tongue up→ B3-1 Spit the tongue right→ B1 Spit the tongue down.
A4	"Slightly open mouth, stick tongue on top of incisor, hold it firmly for ten seconds. Repeat the for ten times."	B6 Tongue on top incisors for 10 Sec

days of training a week; a total of 70 training actions a day, divided into three sessions before meals. Expert E4 suggested taking the training game's themes from older adults' leisure hobbies. Expert E5 confirmed training actions A1~A4 are feasible to be recognized by the TF-Lite model.

2.2 Study 02: User Interviews and Game Theme Design

We recruited five middle-aged and older adults T1 to T5 from local Facebook groups (Figure 2) had a 30-minute semi-structured interview about their life experiences and leisure interests. Three of the five people had heard of "tongue training/exercise." T1 had swallowing difficulties due to stroke and had received training

knowledge from his speech therapist; T2 said she learned about this type of "body maintenance technique" when she borrowed a book about treatment methods. Three people have the habit of exercising. T2 said that the exercise record of her jogging App would encourage her to improve his hiking routine. Almost all of the participants have their way of enjoying music. T1 said that after being discharged from the hospital after a stroke, even though he suffered from speech disorders, he would still try to sing karaoke with friends. T5 shared that he would cut to the music station before bed and play it until he fell asleep. T3 used to ask his family to put music files into his cell phone to enjoy. T4 often uses YouTube to listen to old songs. On the whole, music appreciation has to be found a common interest among the respondents, which cloud

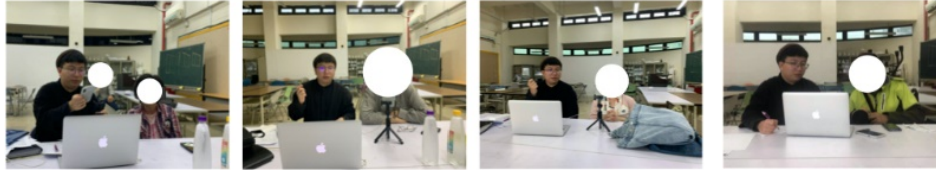


Figure 2: Process of user interviews

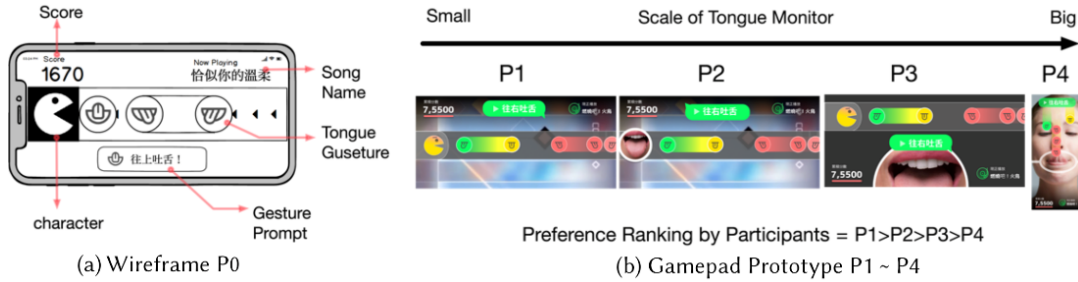


Figure 3: Wireframe and Gamepad Prototype

Table 4: Participant List of Gamepad Prototype Testing

Participant Number	Gender	Age	Preference Sorting
aT1	M	64	P1>P3>P2>P4
aT2	F	62	P1>P2>P3>P4
aT3	M	55	P4>P3>P2>P1
aT4	M	61	P1>P2>P3>P4
aT5	F	55	P1>P2>P3>P4
Average Preference Sorting			P1>P2>P3>P4

be the theme of the exergame; the game should also provide the exercise record feature to help users know their progress.

2.3 Study 03: Gamepad Prototype Testing

In the first iteration, we designed a training game similar to "Taiko no Tatsujin" [17] based on the game theme established in the previous stage. Players need to follow the music beats in the game to make the training action. According to the wireframe (Figure 3 (a)), we designed four kinds of UI layouts P1 ~ P4 (Figure 3 (b)). Five older adults aged 55-64 years old were recruited (Table 4) for the Gamepad method testing [15]. Five expressed a negative attitude towards the P4 interface among the six interviewees. Participants aT1, aT2, aT4, and aT5 told that they did not want to see their full faces on the game screen. aT2 said: "Since I have no makeup in my old age, I feel a bit embarrassed to see my wrinkled face." aT2 and aT3 both had similar opinions about the P3 prototype. aT2 said: "I think it's a bit tiring to look at both the left and the bottom of the screen."; aT3 said: "I have to pay attention to the elf on the left and look at the tongue on the bottom of your screen at the same time. It seems a bit tiring". aT2 said he liked the vivid style and cartoon-like character. aT1 was confused about the direction of the screen: "The screen was vertical when I first logged in, but now it's horizontal, and I'm not used to it."

2.4 Study 04: Wizard of Oz Prototype Testing

The prototype was designed based on the previous evaluation in the second iteration, mixing the P1 Layout and P4 Layout. Training icons become a cartoon-like style, using bright and orange to color the interface. The UI design also considered the interface design guidelines of older adults, including providing clear exit buttons, components larger than 14pt, and the color contrast ratio of text is higher than 1:1.5 [18]. Players can view their training records (Figures 4 (a)), choose training music (Figures 4(b)); In the Game (Figures 4(c)), the training icon will move from top to bottom. Players need to do the specified training action to the camera. When heroes reach the orange area after the game is finished, players can share scores with friends or family (Figures 4(d)(e)).

We used the Wizard of Oz method in the testing; a total of 5 voluntary participants from T1 to T5 (Age = 55~63; M = 59 years old; SD = 3.4 years; 2 males and 3 females) participated in the test, which reached the average level SUS score of 68. All the subjects were first-time users interviewed, informed of the experiment, and gave informed consent before the experiment. The prototype was run on a 5.6-inch smartphone with two cameras next to the subject, observing how the user touched the screen and recording the subject's facial expression (Figures 4(f)). The host guides the user to tap to launch the prototype's APP, login, select a song, and finally

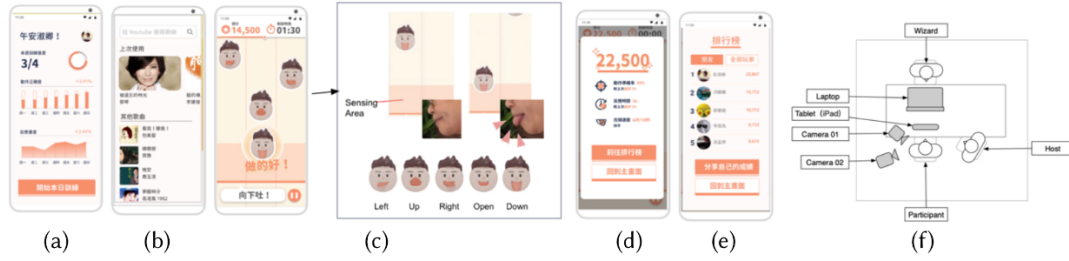


Figure 4: Second Iteration/Wizard of Oz Prototype

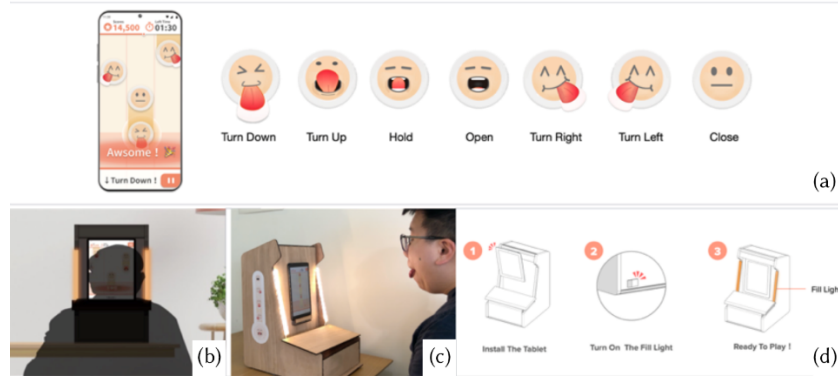


Figure 5: Iteration V2.5 and Arcade Version Design

start the game during the experiment. While the game is playing (Figures 4(c)), the wizard observes the correct mouth movements from the webcam screen, plays the wrong or right sound effects, and manually adds up the scores. After the game, the host guided the participants to fill out the SUS questionnaire and interview them based on their performance to understand their feelings or suggestions about the game. The score of the SUS was 68, T1, T3, T4 said that the speed of the game prototype was too fast, "I think that the speed was too fast, [...] training icon is too small and vague [...] " (T5); T2, T3, and T4 indicated that they liked the concept of using music for training. For example, "I think it's more fun than the original (tongue)gymnastics [...] Can I use other songs I like in this game?" (T2); three participants also mentioned that this design reminded them of "good old days" when they were young. For example, "This is like the dancing machine and dancing mat that I used to play, right?" (T3)

2.5 Arcade Version Design and Icon Revise

After Wizard of Oz Prototype Testing, the icons were redesigned (Figures 5(a)), changing the style from 3D to 2D; the color scheme has also been adjusted, making it easier for users to distinguish between icons. Besides, to meet the needs of residents who live in long-term care facilities or are not familiar with smartphones, we designed a nostalgic arcade version of MusicTongue using wood boxes and a tablet (Figures 5(b)(c)(d)), the way of operation is the same. Still, the Arcade version has a larger screen, is more suitable

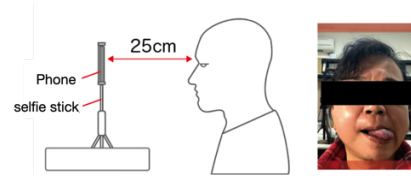


Figure 6: Dataset Image Collection Process

for public space like the parlor of long-term care facilities, generates more social interaction among residents.

3 TECHNICAL APPROACH

We implemented the Image-Recognition model with TF-Lite. All the Dataset images were collected from 10 people aged 55 to 75-year-old, with an Apple iPhone X front camera to take face pictures at a distance of 25 cm (Figure 6). Participants were recruited through an online form on a middle-aged related Facebook group. A total of 4,800 photos were used for image recognition training; Beat-Detection-Unity-library(BDU) [19] generated game stages with the song's beats and length. After the user selects its favorite song, BDU will analyze the rhythms of the music to create a set of metadata; training action A1~A4 will be paired to beats to generate the game stage.



Figure 7: Technical Prototype Testing with PC Version

Table 5: Technical Prototype Test Result

Questionnaire	Result	Interpretation
SUS V3.0 Prototype	79	V3.0 Prototype made an 11-point improvement
UEQ-S Overall	1.91	Higher Than Overall Excellent Level Score 0.92
UEQ-S Hedonic Quality, HQ	1.76	Higher Than HQ Excellent level Score 0.76
UEQ-S Pragmatic Quality, PQ	2.06	Higher Than PQ Excellent level Score 0.91
NPS	19	Lower Than Average NPS Score 31 in "Software & Apps"
Flow Distance, FD	0.19	Between Balance(FD =0) and Boring(FD =1)
NASA-TLX Rounded Score	36.6	Higher Than 75% Video Game Activity[21]

3.1 Technical Prototype

The technical prototype runs on Windows 10 Laptop. During the gameplay (Figure 3 (a)), the player's position accuracy, total score, playtime, and final speed of the game stage will continue to be recorded. According to the accuracy of the player's action, the screen will display three different responses, such as "Keep It Up!", "Good!" and "Awesome!"

4 STUDY 05: TECHNICAL PROTOTYPE TESTING

After implementing the technical prototype, we recruited 36 older adults (Mean = 58.9 years old; SD=7.8; 10 males and 26 females) through the local Facebook group to conduct technical prototype testing (Figure 7(b)). Before the test, the host will lead the participants in a 5-minute practice; the participants will learn how to perform the training exercises and adjust the game's speed through the two-button keyboard. After the session finished, the 30-minute test begins; participants will play three different game levels, fill out the UX questionnaires and conduct in-depth interviews about the game experience.

5 FINDING

The questionnaire results (Table 5) found that SUS scored 79, higher than the previous SUS score of 68 of the second prototype. The overall UEQ-S score is 1.91, with an Excellent level on both Pragmatic Quality (2.06) and Hedonic Quality (1.76) [20]. In the in-depth interview results, almost all of the participants said that the meaning of the icons was easy to understand and recognize while playing, the visibility of the prototype improved after changing the icon style. On the whole, respondents expressed positive attitudes while playing the game but expected more challenging exercises to increase playability; some said they were willing to play with their grandchildren.

6 CONCLUSION

In this study, an oral-training game—MusicTongue, was designed to collaborate with experts and older adults. After three iterations, the game engaged most older adults in oral training. We will be using the crowd-sourcing method to collect more training images from people of different ages, sex, and skin color to increase the diversity of the Training dataset, making the recognition model more accurate and stable. The other training actions in the "Oral Care Program" will also be add-in complete the training process.

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